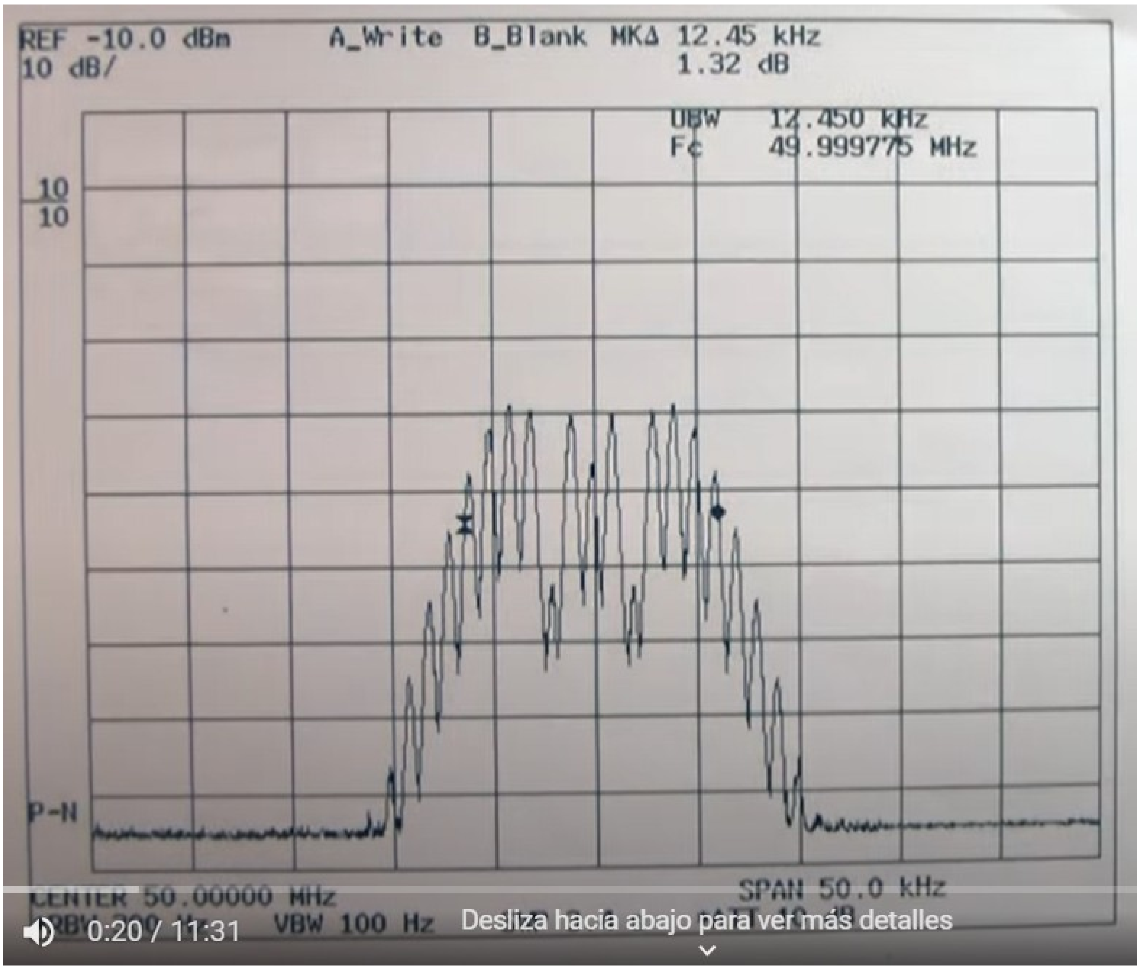
**Actividad espectro de FM**.-En los espectros de FM de las figuras, hallar frecuencia central, frecuencia de la moduladora, índice de modulación, desviación de frecuencia, ancho de banda (regla de Carson), Potencia de la portadora y bandas laterales.




$f_c$ = 50 MHz

$f_m$ = 1 kHz

m = β = 5 (tabla)

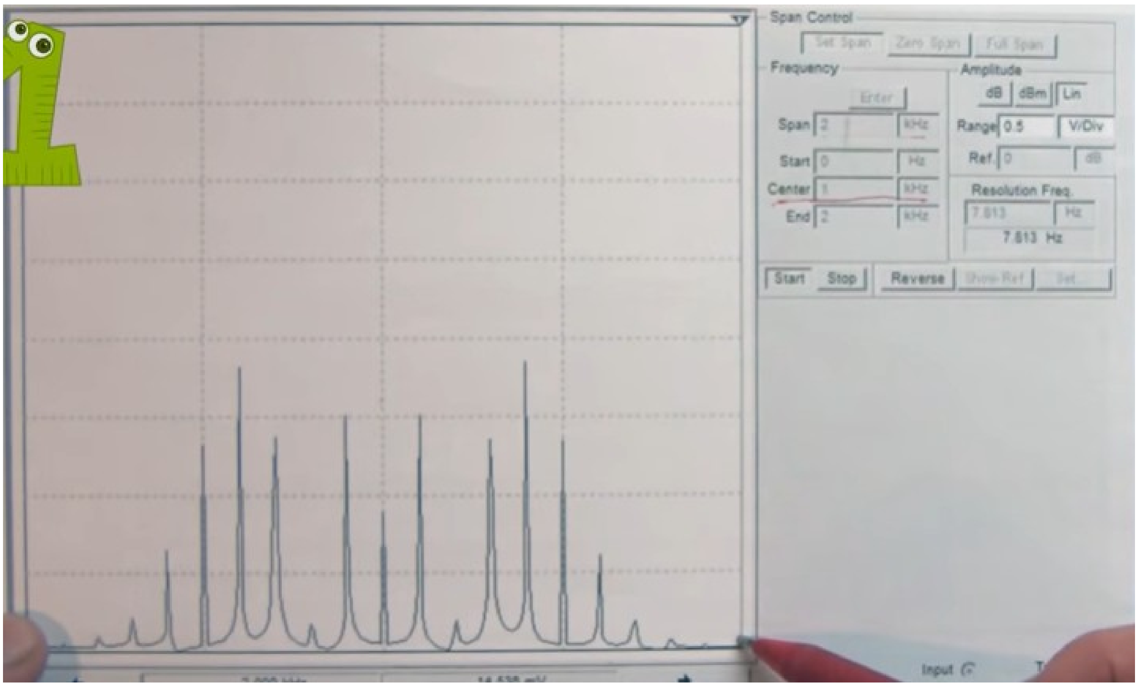
m = β = $\Delta f/f_m$ ; $\Delta f$ = 5 kHz

Carson: BW = $2\cdot(\Delta f+f_m)$ = 12 kHz

…

-10 dBm (referencia, el límite en lo alto) = 0,1 mW

10 dBm por división … $f_c$ = -57 dBm = $10\cdot\log(P/1mW)$ = antilog (-57/10) = (P/1mW) = $1{,}99\times10^{-6}$ mW




**1/**

$f_c$ = 1 kHz

$f_m$ = 100 Hz

m = β = 5

m = β = $\Delta f/f_m$ ; $\Delta f$ = 500 Hz

Carson: BW = $2\cdot(\Delta f+f_m)$ = 1,2 kHz

…

$V_c$ = 0,9 V

$V_1$ = 1,5 V

$V_2$ = 0,2 V

$V_3$ = 1,4 V

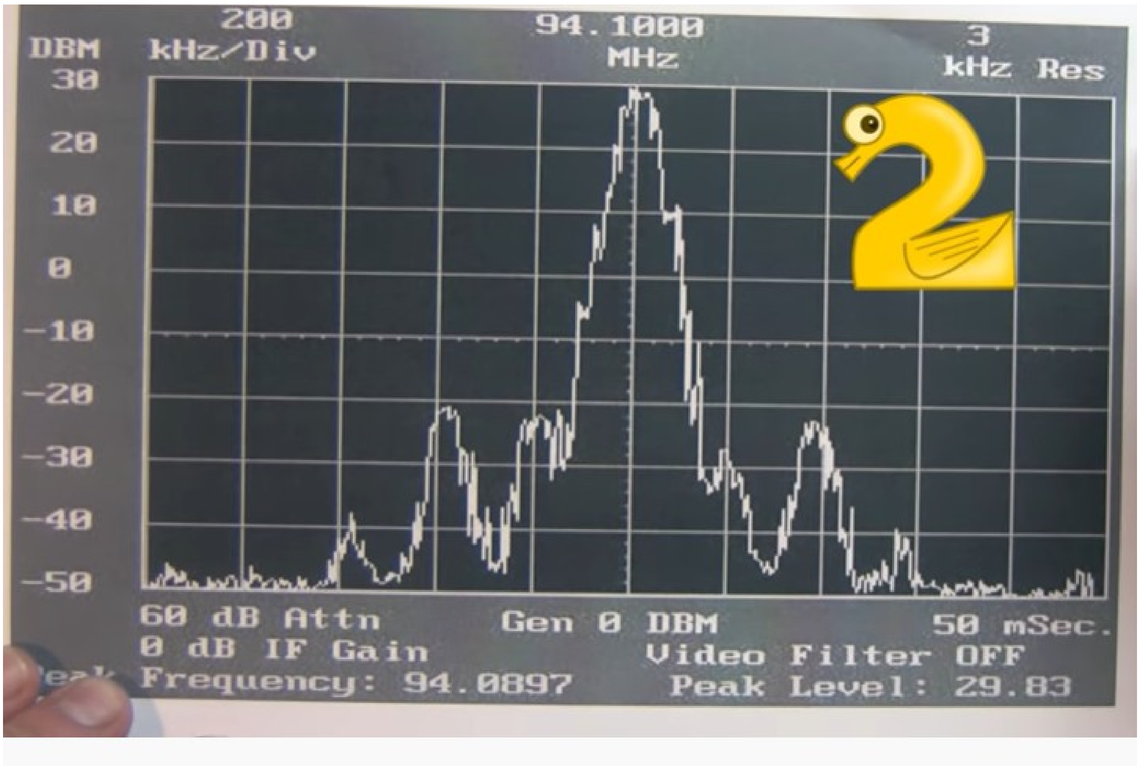
$V_4$ = 1,9 V

$V_5$ = 1,4 V

$V_6$ = 0,6 V

$V_7$ = 0,2 V

$V_8$ = 0,1 V




**2/**

$f_c$ = 94,1 MHz

$f_m$ = 400 kHz

m = β = 0,5

m = β = $\Delta f/f_m$ ; $\Delta f$ = 200 kHz

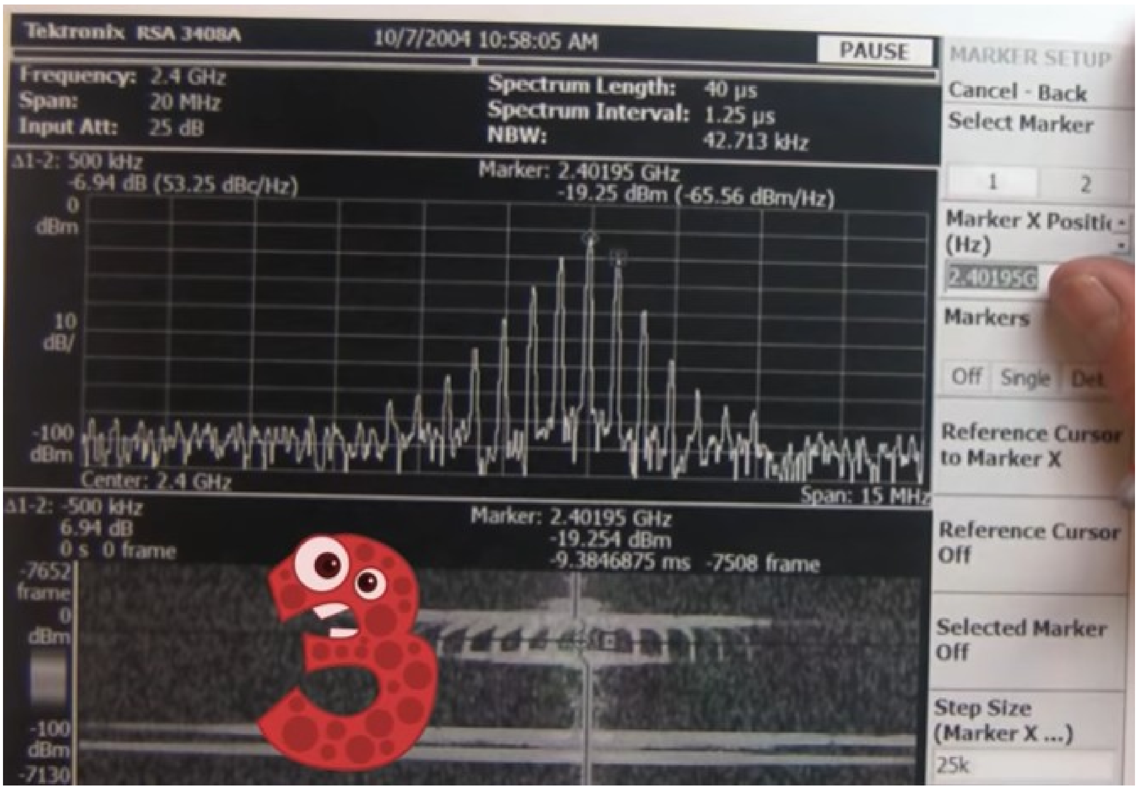
Carson: BW = $2\cdot(\Delta f+f_m)$ = 1,2 kHz

…

30 dBm = 1 W

-20 dBm = 10 µW

-50 dBm = 10 nW




**3/**

$f_c$ = 2,4 GHz

$f_m$ = 500 kHz

m = β = 1

m = β = $\Delta f / f_m$ ; $\Delta f$ = 500 kHz

Carson: BW = $2\cdot(\Delta f + f_m)$ = 2 MHz

…

-10 dBm; $P_c$ = 0,1 mW

-20 dBm; $P_1$ = 0,01 mW

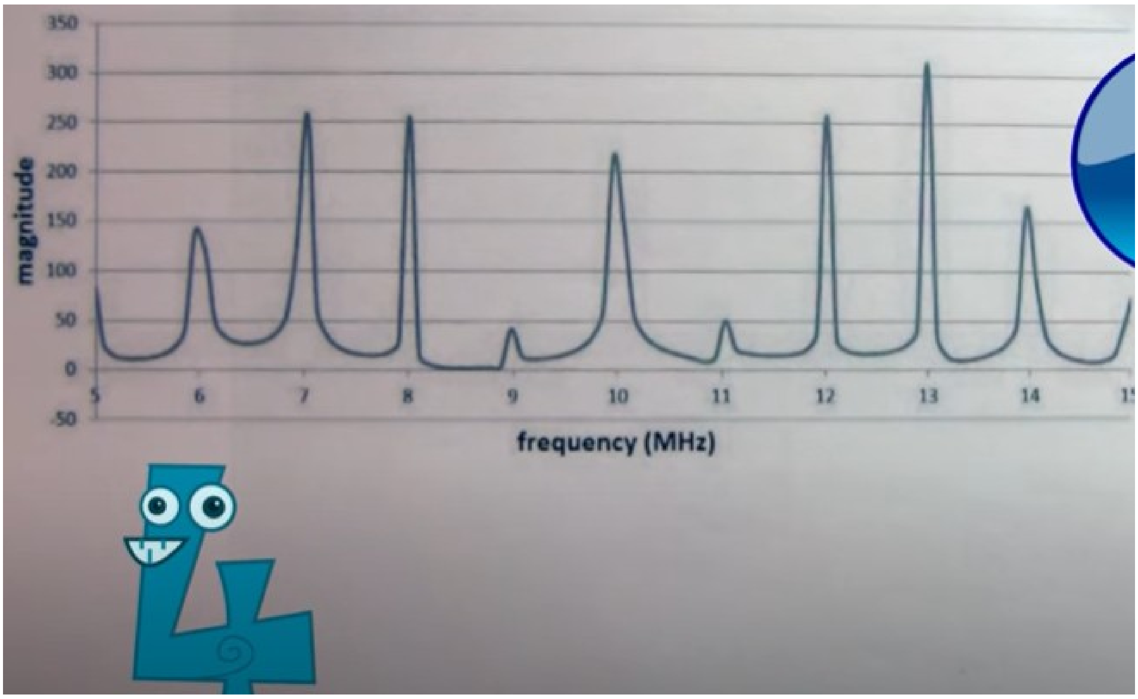
-40 dBm; $P_2$ = 100 nW

-55 dBm; $P_3$ = 3,16 nW

-65 dBm; $P_4$ = 316,23 pW

-70 dBm; $P_5$ = 100 pW

-75 dBm; $P_6$ = 31,62 pW




**4/**

$f_c$ = 10 MHz

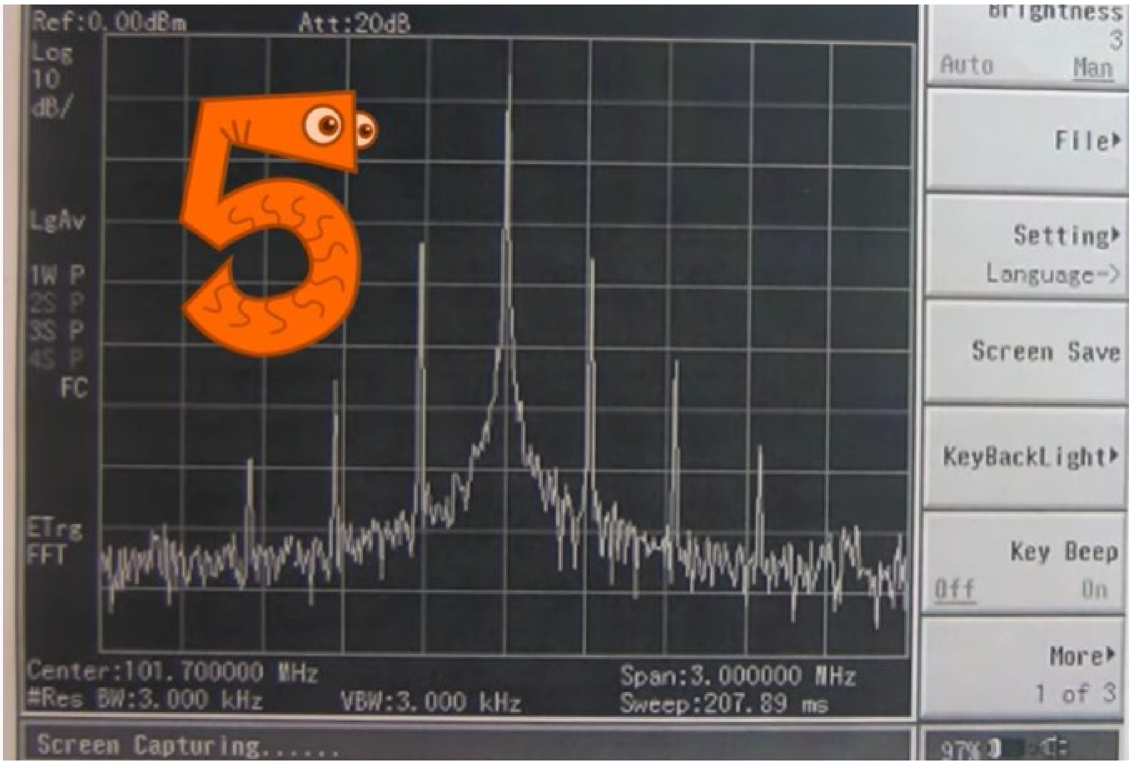
$f_m$ = 1 MHz

m = β = 4

m = β = $\Delta f/f_m$ ; $\Delta f$ = 4 MHz

Carson: BW = $2\cdot(\Delta f+f_m)$ = 10 MHz

…




**5/**

$f_c = 101{,}7$ MHz

$f_m = 0{,}3$ MHz

$m = \beta = 1$

$m = \beta = \Delta f / f_m$ ; $\Delta f = 0{,}3$ MHz

Carson: $BW = 2 \cdot (\Delta f + f_m) = 1{,}2$ MHz

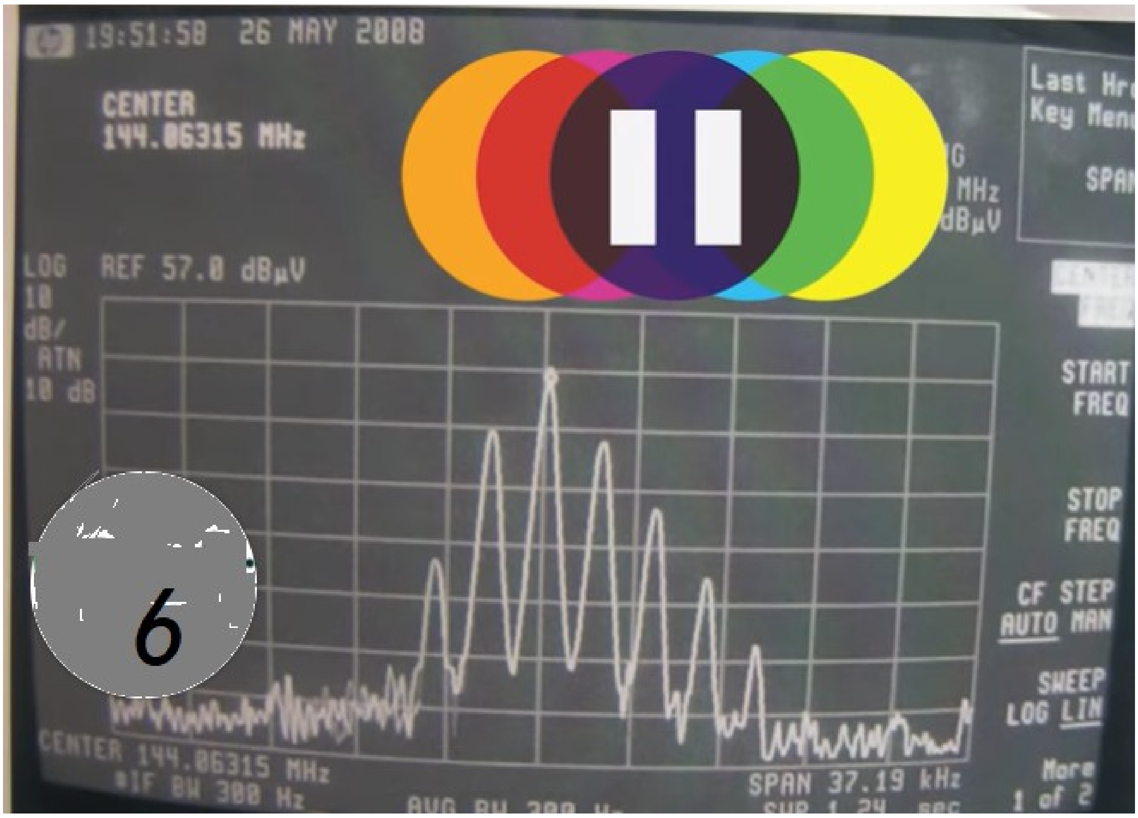
…

-5 dBm; $P_c = 0{,}3$ mW

-35 dBm; $P_1 = 316$ nW

-55 dBm; $P_2 = 3{,}16$ nW

-65 dBm; $P_3 = 316$ pW




**6/**

$f_c$ = 144,86 MHz

$f_m$ = 2,5 kHz

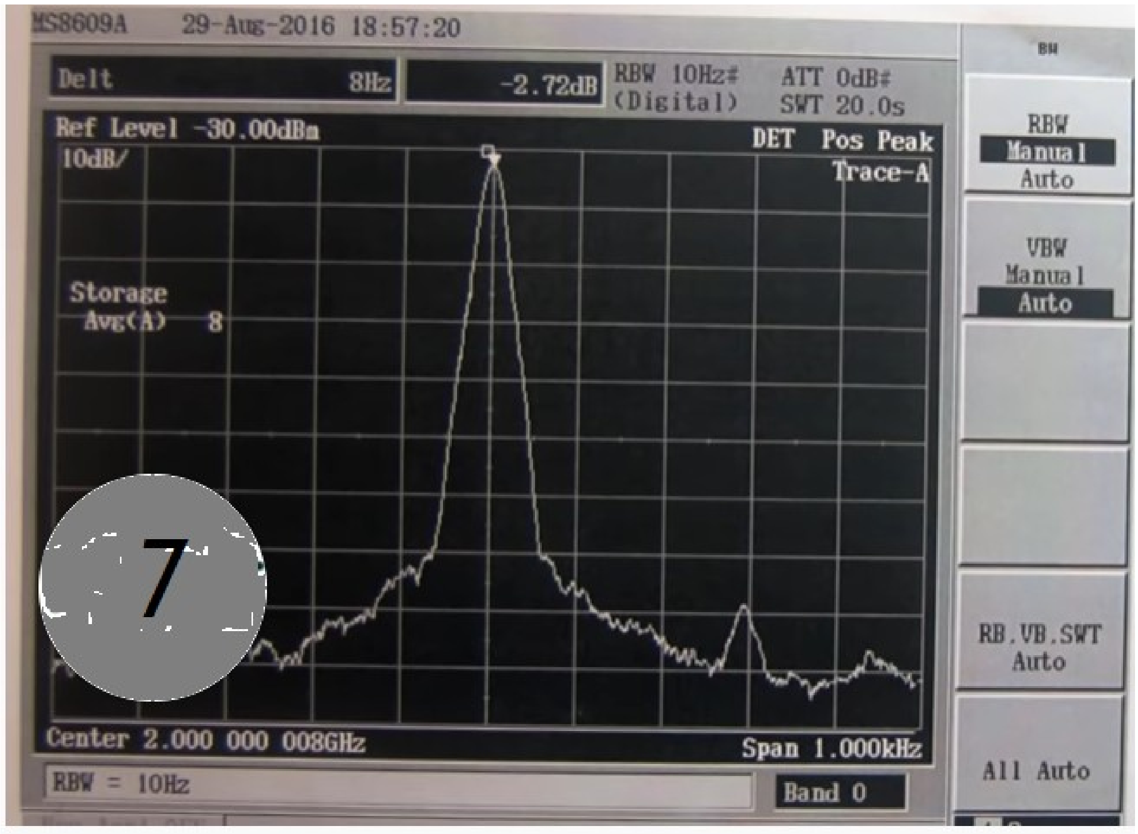
m = $\beta$ = 1

m = $\beta = \Delta f / f_m$ ; $\Delta f$ = 2,5 kHz

Carson: BW = $2 \cdot (\Delta f + f_m)$ = 10 kHz

…

47 dBπ; $P_c$




**7/**

$f_c$ = 2 GHz

$f_m$ = 0,3 kHz

m = β = 0,25

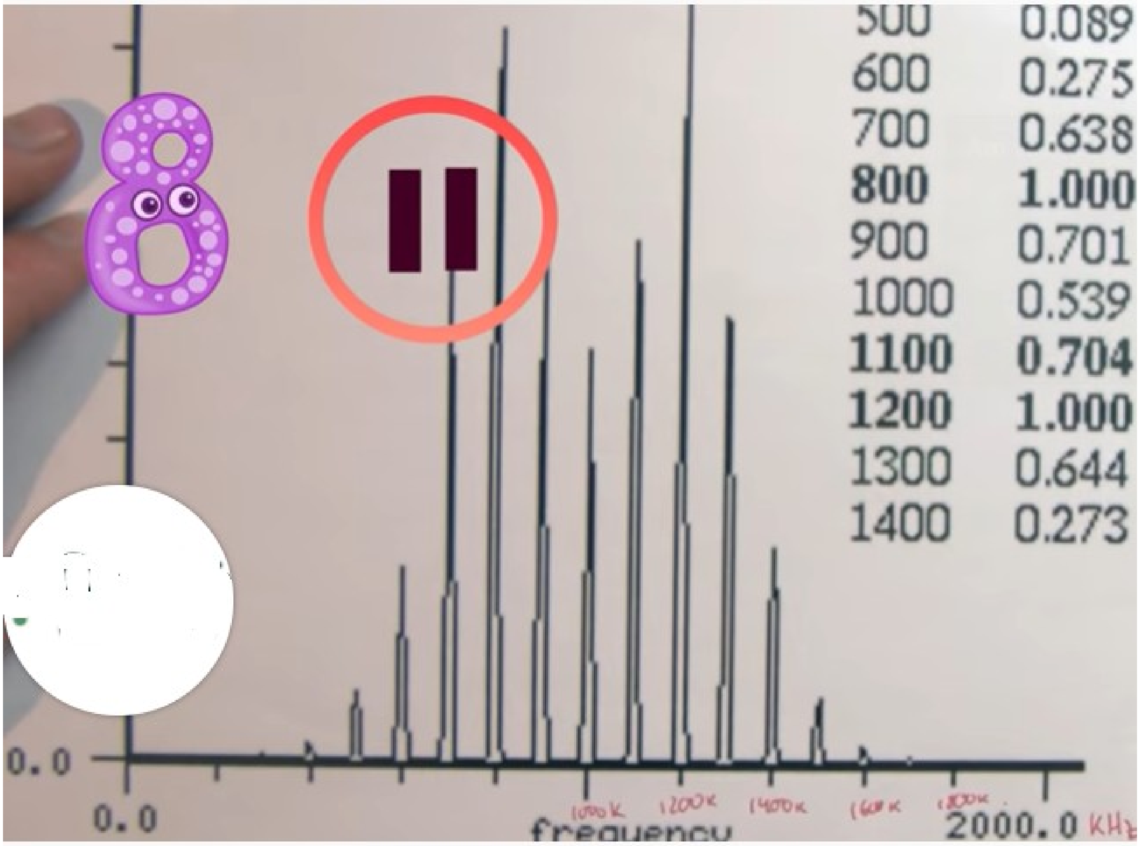
m = β = $\Delta f/f_m$ ; $\Delta f$ = 75 Hz

Carson: BW = $2\cdot(\Delta f+f_m)$ = 750 Hz

…

-30 dBm; $P_c$ = 1 µW

-110 dBm; $P_1$ = 0,01 µW




**8/**

$f_c$ = 1 MHz

$f_m$ = 100 kHz

m = β = 3

m = β = $\Delta f/f_m$ ; $\Delta f$ = 300 kHz

Carson: BW = $2\cdot(\Delta f+f_m)$ = 800 kHz

…